



Kaiser Permanente Pediatric Care (KPPC) Service Line Agreement - Appendicitis/Abdominal Pain Pediatric Patients aged 0-13 years with Possible Appendicitis

Goal:

The goal of Kaiser Permanente Pediatric Care is to ensure that all our pediatric members receive the best care at the right place at the right time.

Purpose:

The purpose of this agreement is to provide our pediatric patients with the highest quality, safest care possible while striving for the highest satisfaction for our members and their families.

KPPC Surgical Guiding Principles:

- It is the goal of KPPC to provide the optimal care for pediatric patients in a timely fashion and closest to the patient's home.
- Transfers should be avoided whenever care can be safely provided at the presenting facility, reserving transfers for tertiary level care that needs to be provided in a tertiary center. In general, appendicitis is not a tertiary level condition.
- Patients 14 years and older can remain at the presenting center, where care can be provided by the surgeon on call.
- Quality is maintained when KP members are cared for at KP facilities by KP physicians.
- To provide optimal care, every transfer requires a direct discussion between the sending and accepting surgeon.

Definitions:

General Medical Center (GMC): Kaiser Permanente Hospital without an inpatient pediatrics floor:

- Baldwin Park, Irvine, Moreno Valley, Panorama City, Riverside, South Bay, West Los Angeles and Woodland Hills

Specialty Medical Center (SMC): Kaiser Permanente Hospital with an inpatient pediatrics floor:

- Anaheim, San Diego

Tertiary Medical Center (TMC): Kaiser Permanente Hospital with both an inpatient pediatrics floor and a pediatric intensive care unit (PICU):

- Los Angeles, Downey, Fontana

Surgeon On Duty (SOD): General Surgeon on call for the designated medical center

Pediatric Transport and Advice Center (PTAC): "Pediatric Call Hub" staffed by an RN and analyst. PTAC connects caller to the appropriate specialist/hospitalist/intensivist for advice and if needed, admission, bed finding and arranging transport starting with one call to **562-658-3980**.

*Antelope Valley, Western Ventura, Kern, and Coachella Valley all rely on contracted hospitals for their pediatric inpatient care. In the infrequent event that these centers are unable to provide the needed care, they would follow the workflow detailed below for the General Medical Centers.

Steps:

Pediatric Patient (0-13 years old) with abdominal pain presents to ED or Pediatric Urgent Care in GMC.

Provider in ER/Urgent Care MD will perform the standard evaluation with any needed labs and x-rays and if they believe that the patient may have appendicitis then they will call the local/GMC SOD to evaluate the patient. If

the decision is made to obtain a CT scan, the order should be placed for a CT of the abdomen and pelvis with IV, not PO, contrast (74177C). If the SOD believes that the patient has appendicitis, they will make arrangements to perform the appendectomy locally (ages 0-5 years, contact PTAC to arrange to speak to the Regional Pediatric Surgeon on call). The patient can be discharged home after the appendectomy.

If, for some reason, such as excessive vomiting, fever, or perforation, the patient cannot be discharged home following the appendectomy, postoperative transfer arrangements can be made by contacting PTAC and speaking to the SOD at the STMC. If it is clear while in the operating room that the patient will need to be transferred postoperatively, PTAC should be called before or during the surgery to start arranging for the transport. This minimizes the time the patient spends in the PACU.

Patients who are perforated and toxic/septic (hypotensive despite 20 ml/ kg normal saline fluid bolus, tachypneic, distended with diffuse marked tenderness) may require expedited preoperative transfer to the ED/peds ward/PICU at the Tertiary Medical Center. This requires a surgeon to surgeon conversation, coordinated by PTAC, who will first confirm bed availability. The accepting surgeon must also confirm operating room availability at the STMC before accepting the patient. The SOD at the GMC should work with the ED to ensure that fluid and antibiotics are given while awaiting transfer. In these cases, the SOD or ED in the GMC may ask PTAC to contact Pediatrics or the Pediatric Intensivist in the PICU to assist with the management and transfer arrangements. In some cases, local source control with removal of the appendix and purulence may be able to be accomplished sooner at the GMC. In this case plans should be made to transfer the patient as soon as possible from the PACU to the STMC.

If the patient will receive surgery at the STMC, the SOD at the STMC should create the case on the STMC grease board to lessen the OR wait time in the STMC. The patient's chart can be found by searching for the chart by MR number (clicking the box "search my region"), importing the chart and creating the case. The operating room at the STMC should also be notified so that space can be allocated based upon the estimated arrival time. This allows more time to pull the case and is preferred by the perioperative team, even if the case is subsequently cancelled.

Note: If needed, there is a Pediatric Protocol for CT scanning which can be requested at the time that the order is placed and this will reduce the radiation dose. **TO EVALUATE FOR PEDIATRIC APPENDICITIS, WE RECOMMEND A CT WITH IV, NOT ORAL CONTRAST** (Order Code 74177C).

SCAL Pediatric Anesthesiology Transfer Workflow:

A pediatric anesthesiologist is required for any pediatric patient less than one year of age or anyone less than 14 years old with an ASA III or above (American Society of Anesthesiology Physical Status Classification System).

If a transfer is considered solely for anesthesia reasons outside of these criteria, then the referring anesthesiologist must contact an anesthesiologist at the accepting center prior to transfer.

Escalation Policy:

If there is a disagreement between the physicians regarding the best option for patient care, escalation is to follow the workflow for EPRP disagreement, and may be initiated by the general surgeon at the GMC, SMC, or TMC or the PTAC or EPRP physician. This escalation should follow policy of the medical centers involved in the disagreement.

In this case, the medical chart should be sent to the COS (Chief of Service) at each of the involved medical centers so that a prompt follow-up and assessment can be performed. The AAMD and AMD at the associated medical centers and then the Regional COS may be asked to assist in resolving any disagreement.

Appendix: Pediatric Appendicitis Score Options

<http://parcscore.com/PARCWizard/wizard.jsessionid=7DA196D188557FEAA3E637B9ABC2BDAE?execution=e1s1>

Or:

From: Madan Samuel: Pediatric appendicitis score, J Pediatr Surg 2002, 37(6): 877-881.

Table 2.

Comparison of the diagnostic weight of the diagnostic indicants between group 1 and group 2

Diagnostic Indicants	Appendicitis (group 1) Diagnostic Weight	Nonappendicitis (group 2) Diagnostic Weight	PValue*	PAS (10)
Cough/percussion tenderness	0.96	0.37	<.001	2
Hopping tenderness	0.96	0.37	<.001	
Anorexia	0.88	0.12	<.001	1
Pyrexia	0.87	0.12	<.001	1
Nausea/Emesis	0.86	0.14	<.001	1
Tenderness in right lower quadrant	0.84	0.37	<.001	2
Leukocytosis†	0.81	0.2	<.001	1
Polymorphonuclear neutrophilia	0.80	0.22	<.001	1
Migration of pain	0.80	0.2	<.001	1

*P <.05 statistically significant. †Leukocytosis with a white blood count of $\geq 10,000(10^9/L)$.

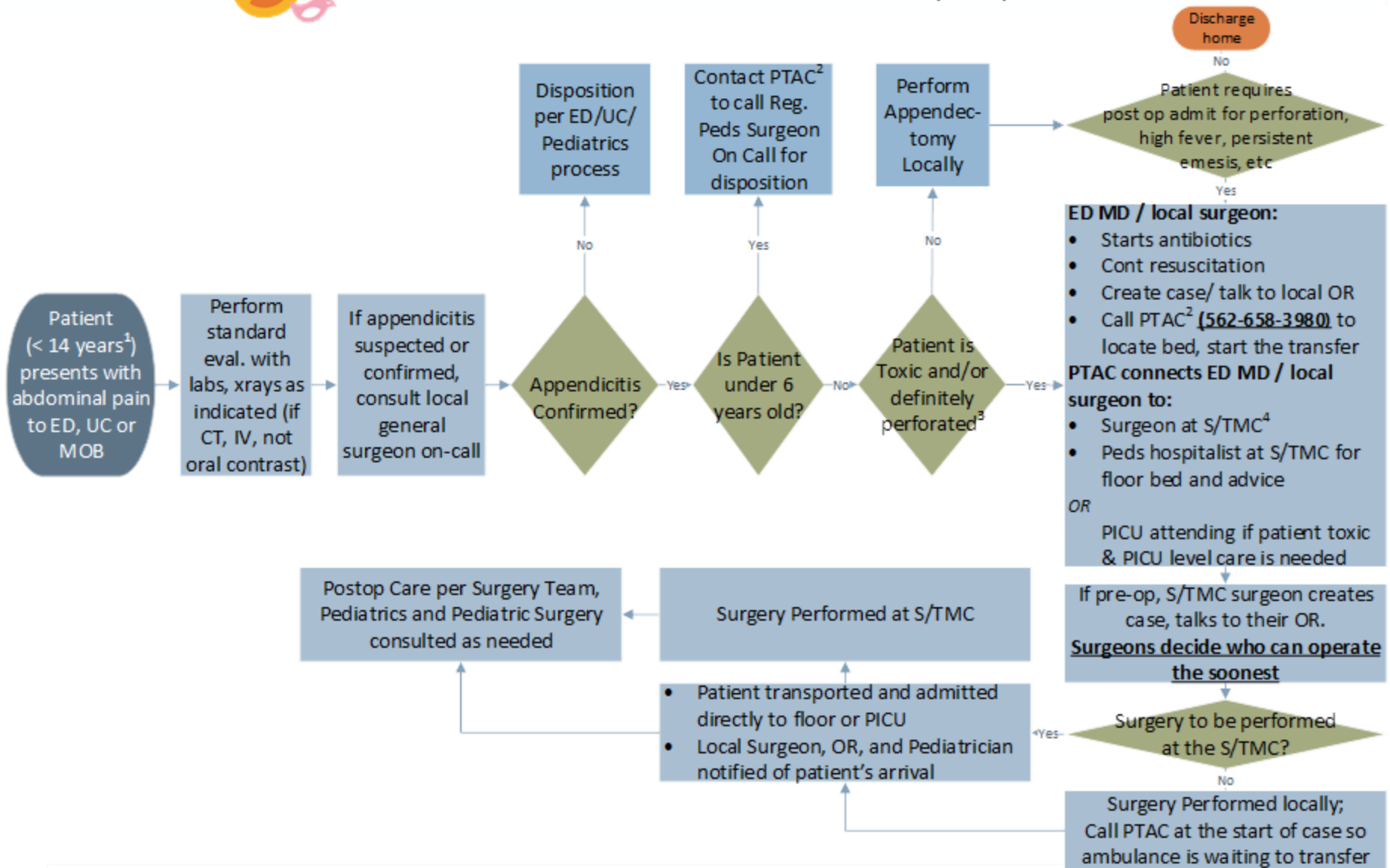
Score of 7 or higher is highly suspicious for appendicitis

Score below 5 is unlikely to be appendicitis.

Examination by an experienced surgeon is highly valuable because no scoring system is perfect



KPPC APPENDECTOMY SURGICAL TRANSFER FLOW: FROM GENERAL TO SPECIALTY/TERTIARY MEDICAL CENTERS (S/TMC)



- Patients from age 14-17 will stay at the center in which they present unless there are clinical conditions present requiring higher level of care.
- PTAC – Pediatric Transport and Advice Center can be reached at **562-658-3980** and should be involved as much as possible for tracking.
- Toxic/septic defined as: hypotensive despite 20 ml/ kg normal saline fluid bolus, tachypneic, distended with diffuse marked tenderness.
- Note: S/TMC – Specialty / Tertiary Medical Center